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Exam first trimester

Status	Finished
Started	Monday, 1 December 2025, 3:06 PM
Completed	Monday, 1 December 2025, 3:38 PM
Duration	32 mins 47 secs
Marks	7.50/55.00
Grade	1.36 out of 10.00 (13.64%)

Question 1

Incorrect

Mark -0.33 out of 1.00

Flag question

What is the purpose of using **predicate** in lambda expressions?

- ☐ a. To define a sorting order
- ☐ b. To perform a conditional check
- ☒ c. To transform data ✖
- ☐ d. To create new objects

The correct answer is: To perform a conditional check

Question 2

Incorrect

Mark -0.33 out of 1.00

Flag question

What is the purpose of the **map** operation in streams?

- ☐ a. To perform a reduction
- ☐ b. To transform each element into another form
- ☒ c. To filter elements based on a condition ✖
- ☐ d. To sort elements

The correct answer is: To transform each element into another form

Question 3

Incorrect

Mark -0.33 out of 1.00

Flag question

Which operation type in streams is responsible for transforming one stream into another?

- ☒ a. Collecting operations ✖
- ☐ b. Reduction operations
- ☐ c. Intermediary or lazy operations
- ☐ d. Final operations

The correct answer is: Intermediary or lazy operations

Question 4

Incorrect

Mark -0.33 out of 1.00

Flag question

What is the primary use of the **collect** method in streams?

- ☒ a. To filter elements ✖
- ☐ b. To finalize a stream and return a collection
- ☐ c. To perform intermediate operations
- ☐ d. To map values

The correct answer is: To finalize a stream and return a collection

Question 5

Incorrect

Mark -0.33 out of 1.00

Flag question

What is a stream in Java?

- ☐ a. A method for direct data manipulation
- ☐ b. A special object for defining operations using lambda expressions
- ☐ c. A type of collection
- ☒ d. A data structure similar to a list ✖

Quiz navigation

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Show one page at a time

The correct answer is: A special object for defining operations using **lambda expressions**

Question 6

Incorrect

Mark -0.33 out of 1.00

Flag question

Which functional interface is used in filter operations?

- ☒ a. Function ☒
- ☐ b. Predicate
- ☐ c. Consumer
- ☐ d. Supplier

The correct answer is: Predicate

Question 7

Incorrect

Mark -0.33 out of 1.00

Flag question

In Java, what is the difference between synchronized collections and concurrent collections?

- ☐ a. Synchronized collections offer better performance
- ☐ b. There is no significant difference
- ☒ c. Concurrent collections block the entire collection on access ☒
- ☐ d. Synchronized collections block the entire collection on access

The correct answer is: Synchronized collections block the entire collection on access

Question 8

Incorrect

Mark -0.33 out of 1.00

Flag question

How can the **synchronized** keyword help in concurrent programming in Java?

- ☒ a. It automatically manages thread priorities ☒
- ☐ b. It improves the performance of the application
- ☐ c. It allows multiple threads to access the method or block simultaneously
- ☐ d. It restricts access to a method or block to one thread at a time

The correct answer is: It restricts access to a method or block to one thread at a time

Question 9

Incorrect

Mark -0.33 out of 1.00

Flag question

What is the primary benefit of using atomic data types for arrays in Java?

- ☐ a. They allow arrays to be shared across multiple virtual machines
- ☐ b. They ensure each operation on the array elements is atomic
- ☒ c. They improve the execution speed of array operations ☒
- ☐ d. They can store larger arrays

The correct answer is: They ensure each operation on the array elements is atomic

Question 10

Correct

Mark 1.00 out of 1.00

Flag question

In thread management, what is a daemon thread?

- ☐ a. A thread with the highest priority
- ☐ b. A thread that cannot be stopped once started
- ☐ c. A thread that manages other threads
- ☒ d. A thread that runs in the background with low priority ✓

The correct answer is: A thread that runs in the background with low priority

Question 11

Incorrect

Mark -0.33 out of 1.00

Flag question

In Java, what is a thread?

- ☒ a. A separate process independent of the main application ☒
- ☐ b. A subprocess or subtask with a shared context with other threads of the same application
- ☐ c. A method within the main application
- ☐ d. A type of memory allocation

The correct answer is: A subprocess or subtask with a shared context with other threads of the same application

Question 12

Incorrect

Mark -0.33 out of 1.00

Flag question

How can a thread's execution be interrupted in Java?

- ☐ a. Setting the thread's priority to the lowest level
- ☐ b. Calling the interrupt() method on the thread
- ☐ c. Removing all references to the thread
- ☒ d. Using the stop() method ❌

The correct answer is: Calling the interrupt() method on the thread

Question 13

Incorrect

Mark -0.33 out of 1.00

Flag question

Which Java class is used to define a thread by inheritance?

- ☐ a. Executor
- ☐ b. Thread
- ☒ c. Runnable ❌
- ☐ d. Callable

The correct answer is: Thread

Question 14

Incorrect

Mark -0.33 out of 1.00

Flag question

How do you properly start a thread in Java?

- ☐ a. Using the execute() method
- ☒ b. Triggering the run() method indirectly ❌
- ☐ c. Calling the run() method directly
- ☐ d. Invoking the start() method

The correct answer is: Invoking the start() method

Question 15

Correct

Mark 1.00 out of 1.00

Flag question

How does the **Lock** interface improve synchronization in Java?

- ☒ a. By providing more flexible locking mechanisms than **synchronized** blocks ✅
- ☐ b. By automatically managing thread pools
- ☐ c. By reducing the overhead of thread creation
- ☐ d. By allowing thread priorities to be changed dynamically

The correct answer is: By providing more flexible locking mechanisms than **synchronized** blocks

Question 16

Incorrect

Mark -0.33 out of 1.00

Flag question

What does the **Future** interface represent in Java concurrency?

- ☐ a. The result of an asynchronous computation
- ☐ b. A lock for synchronizing threads
- ☒ c. A thread's state in the future ❌
- ☐ d. A scheduled task to be executed in the future

The correct answer is: The result of an asynchronous computation

Question 17

Incorrect

Mark -0.33 out of 1.00

Flag question

What is the primary advantage of using a **CompletableFuture** in Java?

- ☒ a. It ensures thread safety without explicit locks ❌
- ☐ b. It simplifies error handling in multithreading
- ☐ c. It automatically cancels long-running tasks
- ☐ d. It allows for non-blocking asynchronous task execution

The correct answer is: It allows for non-blocking asynchronous task execution

Question 18

Incorrect

Mark -0.33 out of 1.00

Flag question

In the context of Java's concurrent programming, what is a `ScheduledExecutorService` used for?

- ☐ a. Running tasks at a specific time
- ☐ b. Scheduling thread termination
- ☒ c. Managing thread priorities ❌
- ☐ d. Synchronizing thread execution

The correct answer is: Running tasks at a specific time

Question 19

Correct

Mark 1.00 out of 1.00

Flag question

Which interface can be implemented as an alternative to extending the `Thread` class for creating a thread?

- ☐ a. `ActionListener`
- ☐ b. `Executor`
- ☐ c. `Callable`
- ☒ d. `Runnable` ✔️

The correct answer is: `Runnable`

Question 20

Correct

Mark 1.00 out of 1.00

Flag question

What is the main difference between extending the `Thread` class and implementing the `Runnable` interface?

- ☒ a. Implementing `Runnable` allows multiple inheritance ✔️
- ☐ b. There is no practical difference
- ☐ c. `Runnable` is more memory-efficient
- ☐ d. Extending `Thread` allows multiple inheritance

The correct answer is: Implementing `Runnable` allows multiple inheritance

Question 21

Correct

Mark 1.00 out of 1.00

Flag question

A lambda expression is used to implement a functional interface

Select one:

- ☒ True ✔️
- ☐ False

The correct answer is 'True'.

Question 22

Correct

Mark 1.00 out of 1.00

Flag question

What will be the output of the program?

```
class MyThread extends Thread
{
    public MyThread() {}
    public MyThread(Runnable r) {super(r); }
    public void run()
    {
        System.out.print("Inside Thread ");
    }
}
class MyRunnable implements Runnable
{
    public void run()
    {
        System.out.print(" Inside Runnable");
    }
}
class Test
{
    public static void main(String[] args)
    {
        new MyThread().start();
        new Thread(new MyRunnable()).start();
    }
}
```

- ☐ a. Does not compile
- ☒ b. Prints "Inside Thread" and "Inside Runnable" in any order ✔️

- ☐ c. Prints "Inside Thread Inside Thread"
- ☐ d. Throws exception at runtime

The correct answer is: Prints "Inside Thread" and "Inside Runnable" in any order

Question 23

Correct

Mark 1.00 out of 1.00

Flag question

This code ...

```
String line;
try (BufferedReader bfRead = new BufferedReader(new
    FileReader("/home/pepe/file.txt")))
{
    line = bfRead.readLine();
    while(line != null)
    {
        line = bfRead.readLine();
    }
} catch (FileNotFoundException e) {
    System.err.println("File /home/pepe/file.txt doesn't exist!");
} catch (IOException e) {
    System.err.println("Error reading /home/pepe/file.txt.");
} finally {
    if(bfRead != null)
    try
    {
        bfRead.close();
    } catch (IOException e) {
        System.err.println("Couldn't close C:\file.txt stream.");
    }
}
```

- ☐ a. has an exception not handled
- ☐ b. doesn't compile because we have to close de file in the first try
- ☒ c. doesn't compile because bfRead is not defined in the finally clause ✓
- ☐ d. is correct

The correct answer is:
doesn't compile because bfRead is not defined in the finally clause

Question 24

Correct

Mark 1.00 out of 1.00

Flag question

What will be the output of the program?

```
class A
{
    public int GetResult(int a, int b) { return 0; }
}
class B extends A
{
    public int GetResult(int a, int b) {return 1; }
}
public class Test
{
    public static void main(String args[])
    {
        B b = new B();
        System.out.println("x = " + b.GetResult(0, 1));
    }
}
```

- ☐ a. Compilation fails.
- ☐ b. x = 0
- ☐ c. An exception is thrown at runtime.
- ☒ d. x = 1 ✓

The correct answer is: x = 1

Question 25

Correct

Mark 1.00 out of 1.00

Flag question

Lazy or intermediary operations in streams are...

- ☐ a. Operations that finish the stream producing a final result
- ☒ b. Operations that produce an additional stream to join additional intermediary operations ✓
- ☐ c. Operations that reduce the stream to a single value
- ☐ d. Operations that produce a List as a result

The correct answer is:

Operations that produce an additional stream to join additional intermediary operations

Question 26

Not answered

Marked out of 1.00

Flag question

This lambda expression

```
Comparator<Integer> comp = (i1, i2) -> Integer.compare(i1, i2);
```

can be written in a shorter way which is:

Answer: ❌

The correct answer is: `Comparator<Integer> comp = Integer::compare;`

Question 27

Incorrect

Mark -0.25 out of 1.00

Flag question

```
class X implements Runnable
{
    public static void main(String args[])
    {
        /* Missing code? */
    }
    public void run() {}
}
```

Which of the following line of code is suitable to start a thread ?

- ☐ a. `X run = new X(); Thread t = new Thread(run); t.start();`
- ☐ b. `Thread t = new Thread(X);`
- ☐ c. `Thread t = new Thread(); x.run();`
- ☒ d. `Thread t = new Thread(X); t.start();` ❌

The correct answer is: `X run = new X(); Thread t = new Thread(run); t.start();`

Question 28

Incorrect

Mark -0.25 out of 1.00

Flag question

If we can link the output of a function with the input of next function, then we are applying...

- ☐ a. Function composition
- ☐ b. Data immutability
- ☒ c. Referential transparency ❌
- ☐ d. First order functions

The correct answer is: Function composition

Question 29

Incorrect

Mark -0.33 out of 1.00

Flag question

Runnable is

- ☐ a. An interface that force to override a method called run and another called start
- ☐ b. a functional interface that allows to create Threads
- ☐ c. An interface to create new Callables
- ☒ d. An abstract class to create new Threads ❌

The correct answer is:
a functional interface that allows to create Threads

Question 30

Incorrect

Mark -0.25 out of 1.00

Flag question

This class...

```
public class UpperCaseReader
{
    public UpperCaseReader(Reader reader)
    {
        super(reader);
    }

    @Override
    public String readLine() throws IOException
    {
        String line = super.readLine();
        return line != null ? line.toUpperCase() : null;
    }
}
```

- ☐ a. will be correct if we add "extends BufferedReader" in the first line
- ☐ b. is correct and allows us to read a file only in upperCase
- ☒ c. will be correct if we remove the override annotation ❌

☐ d. will be correct if we remove the "throws" instruction

The correct answer is:
will be correct if we add "extends BufferedReader" in the first line

Question 31

Incorrect

Mark -0.25 out of 1.00

Flag question

Which will be the output for this piece of code?

```
import java.util.*;

public class Test
{
    public static void main(String[] args)
    {
        ArrayList<String> texts = new ArrayList<>();

        texts.add("aaa");
        texts.add("bbb");

        texts.stream().filter(s -> s.startsWith("a"))
            .map(s -> s + " mapped")
            .forEach(s -> System.out.println(s));
    }
}
```

- ☐ a. aaa
- ☐ b. aaa mapped
- ☐ c. mapped
- ☒ d. Compilation error ❌

The correct answer is:
aaa mapped

Question 32

Incorrect

Mark -0.25 out of 1.00

Flag question

This code...

```
List<String> list = Arrays.asList("Hello", "Hi", "Goodbye", "Farewell", "Bye");
Collections.sort(list, (s1,s2) -> Integer.compare(s1.length(), s2.length()));
```

- ☐ a. is incorrect
- ☐ b. implements a functional interface to sort a list
- ☐ c. uses an anonymous class to sort a list
- ☒ d. uses a normal class to implement the Comparator interface ❌

The correct answer is:
implements a functional interface to sort a list

Question 33

Not answered

Marked out of 1.00

Flag question

In this code:

```
future=executor.submit(Addition(1,2,3));
```

Addition is a

Answer: ❌

The correct answer is: Callable

Question 34

Incorrect

Mark -0.25 out of 1.00

Flag question

In this instruction:

```
executor.execute(singleEncryption(f));
```

singleEncryption is

- ☐ a. A Callable
- ☐ b. A future
- ☐ c. A Runnable
- ☒ d. None is correct ❌

The correct answer is:
A Runnable

Question 35

Incorrect

Mark -0.25 out of 1.00

Flag question

```
class A
{
    A( ) { }
}
class B extends A
{ }
```

Which statement is true?

- ☐ a. Class B'S constructor has no arguments.
- ☒ b. None of these.
- ☐ c. Class B'S constructor is private.
- ☐ d. Class B'S constructor includes a call to this().

The correct answer is: Class B'S constructor has no arguments.

Question 36

Correct

Mark 1.00 out of 1.00

Flag question

In JavaFX applications, BorderPane is a layout container that lets us arrange the controls in 5 different areas: TOP, BOTTOM, LEFT, RIGHT and CENTER.

Select one:

- ☒ True
- ☐ False

The correct answer is 'True'.

Question 37

Partially correct

Mark 0.38 out of 1.00

Flag question

If we want to make sure that an integer variable is going to be properly modified by a set of 1000 threads, we need to... (mark every possible answer)

- ☒ a. Use an executor to make sure that every thread accesses the variable in a coordinated way
- ☒ b. Synchronize every possible method that modifies the value of this variable using "synchronized" word
- ☐ c. Create the threads through Runnable interface instead of extending Thread class
- ☐ d. Use an AtomicInteger variable to make sure that the integer value is thread-safe

The correct answers are:

Synchronize every possible method that modifies the value of this variable using "synchronized" word,

Use an AtomicInteger variable to make sure that the integer value is thread-safe

Question 38

Correct

Mark 1.00 out of 1.00

Flag question

According to the following piece of code...

```
Thread t = new Thread() -> {
    for (int i = 1; i <= 1000; i++)
        System.out.println(i);
};

t.sleep(5000);
t.start();
System.out.println("Finished");
```

- ☐ a. Main program will sleep for 5 seconds, then it will print "Finished" message and then it will launch thread t
- ☒ b. Main program will sleep for 5 seconds, then it will start thread t and finally it will print "Finished" message
- ☐ c. Thread t will sleep for 5 seconds, then it will print "Finished" message and then it will launch thread t
- ☐ d. Thread t will sleep for 5 seconds, and then it will start counting from 1 to 1000

The correct answer is:

Main program will sleep for 5 seconds, then it will start thread t and finally it will print "Finished" message

Question 39

Incorrect

Mark -0.25 out of 1.00

Flag question

What will be the output of the program?

```
try
{
    int x = 0;
    int y = 5 / x;
}
```

```

    catch (Exception e)
    {
        System.out.println("Exception");
    }
    catch (ArithmeticException ae)
    {
        System.out.println(" Arithmetic Exception");
    }
    System.out.println("finished");

```

- ☒ a. Arithmetic Exception ❌
- ☐ b. Compilation fails because Exception already contains ArithmeticException
- ☐ c. Exception
- ☐ d. finished

The correct answer is: Compilation fails because Exception already contains ArithmeticException

Question 40

Incorrect

Mark -0.25 out of 1.00

Flag question

What will be the output of the program?

```

class MyThread extends Thread
{
    public static void main(String [] args)
    {
        MyThread t = new MyThread();
        t.start();
        System.out.print("one. ");
        t.start();
        System.out.print("two. ");
    }
    public void run()
    {
        System.out.print("Thread ");
    }
}

```

- ☒ a. The output cannot be determined. ❌
- ☐ b. An exception occurs at runtime.
- ☐ c. It prints "Thread one. Thread two."
- ☐ d. Compilation fails

The correct answer is: An exception occurs at runtime.

Question 41

Correct

Mark 1.00 out of 1.00

Flag question

A Thread can be created inheriting from class Thread or implementing the functional interface ...

Answer: Runnable ✅

The correct answer is: Runnable

Question 42

Not answered

Marked out of 1.00

Flag question

Which statement is true for the class java.util.ArrayList?

- ☐ a. The collection is guaranteed to be immutable.
- ☐ b. The elements in the collection are accessed using a unique key.
- ☐ c. The elements in the collection are ordered/indexed.
- ☐ d. The elements in the collection are guaranteed to be unique.

The correct answer is: The elements in the collection are ordered/indexed.

Question 43

Correct

Mark 1.00 out of 1.00

Flag question

Anonymous classes are known in many languages as lambda expressions

Select one:

- ☐ True
- ☒ False ✅

The correct answer is 'False'.

Question 44

What will be the output of the program?

Incorrect

Mark -0.25 out of 1.00

Flag question

```
What will be the output of the program:  
class s implements Runnable  
{  
    int x, y;  
    public void run()  
    {  
        for(int i = 0; i < 1000; i++)  
            synchronized(this)  
            {  
                x = 12;  
                y = 12;  
            }  
        System.out.print(x + " " + y + " ");  
    }  
    public static void main(String args[])  
    {  
        s run = new s();  
        Thread t1 = new Thread(run);  
        Thread t2 = new Thread(run);  
        t1.start();  
        t2.start();  
    }  
}
```

- ☐ a. It prints 12 12 12 12
- ☒ b. Compilation Error ❌
- ☐ c. Cannot determine output.
- ☐ d. DeadLock

The correct answer is: It prints 12 12 12 12

Question 45

Correct

Mark 1.00 out of 1.00

Flag question

A stream is a collection of *java.util*

Select one:

- ☐ True
- ☒ False ✅

The correct answer is 'False'.

Question 46

Incorrect

Mark 0.00 out of 1.00

Flag question

The difference between Callable and Runnable is that Callable is void and Runnable return a value.

Select one:

- ☒ True ❌
- ☐ False

The correct answer is 'False'.

Question 47

Not answered

Marked out of 1.00

Flag question

If a functional interface has one method to implement with just one parameter...

- ☐ a. We can omit the parentheses at the left part of the arrow when defining the lambda expression
- ☐ b. We should use an anonymous class instead of a lambda expression
- ☐ c. We can't implement it using a lambda expression
- ☐ d. We can't inherit from that interface

The correct answer is: We can omit the parentheses at the left part of the arrow when defining the lambda expression

Question 48

Partially correct

Mark 0.88 out of 1.00

Flag question

Look at this piece of code

```
public void myFunction()  
{  
    anotherFunction()  
}
```

If "anotherFunction" can throw an exception, and we get a compilation error because of this, we can solve this issue by... (mark every possible solution)

- ☒ a. Declare this exception to be thrown in "myFunction" using "throws" clause ✅
- ☒ b. Catch the possible exception when calling "anotherFunction" with the corresponding "try" clause ✅

- ☒ c. Throw the exception manually inside "myFunction" with "throw" ❌
- ☐ d. Use "finally" to avoid catching the possible exception

The correct answers are:

Declare this exception to be thrown in "myFunction" using "throws" clause,

Catch the possible exception when calling "anotherFunction" with the corresponding "try" clause

Question 49

Incorrect

Mark -0.25 out of 1.00

Flag question

In this piece of code where wordCounter returns an int:

```
futures = executor.invokeAll(Arrays.asList(
    wordCounter(word, "text1.txt"),
    wordCounter(word, "text2.txt"),
    wordCounter(word, "text3.txt")
));
```

executor.shutdown();

futures is ...

- ☐ a. List<Integer>
- ☒ b. int ❌
- ☐ c. List<Callable<Integer>>
- ☐ d. List<Future<Integer>>

The correct answer is:

List<Future<Integer>>

Question 50

Correct

Mark 1.00 out of 1.00

Flag question

Which of the following is an example of a functional interface in Java?

- ☐ a. ArrayList
- ☐ b. String
- ☒ c. Comparator ✔️
- ☐ d. HashMap

The correct answer is: Comparator

Question 51

Incorrect

Mark -0.33 out of 1.00

Flag question

How can lambda expressions simplify code in Java?

- ☐ a. By automatically generating method names
- ☐ b. By implementing interfaces without creating a class
- ☐ c. By replacing all anonymous classes
- ☒ d. By removing the need for loops ❌

The correct answer is: By implementing interfaces without creating a class

Question 52

Incorrect

Mark -0.33 out of 1.00

Flag question

How do lambda expressions enhance the functionality of collections in Java?

- ☐ a. By allowing immutable collections
- ☒ b. By automatically sorting collections ❌
- ☐ c. By simplifying iteration and manipulation
- ☐ d. By introducing parallel processing

The correct answer is: By simplifying iteration and manipulation

Question 53

Incorrect

Mark -0.33 out of 1.00

Flag question

Which of these is the correct syntax for a lambda expression in Java?

- ☐ a. () -> {}
- ☒ b. {() -> } ❌
- ☐ c. () => {}
- ☐ d. {}() => }

The correct answer is: `() -> {}`

Question 54

Incorrect

Mark -0.33 out of 1.00

Flag question

Which of the following is true about lambda expressions?

- ☐ a. They require functional interfaces
- ☒ b. They replace the need for interfaces ❌
- ☐ c. They can throw checked exceptions without declaration
- ☐ d. They are used to implement methods from multiple interfaces

The correct answer is: They require functional interfaces

Question 55

Correct

Mark 1.00 out of 1.00

Flag question

In a lambda expression, what does the arrow (->) symbolize?

- ☐ a. A method call
- ☐ b. Direction of data flow
- ☒ c. Separation between parameters and implementation ✔️
- ☐ d. Pointer to an object

The correct answer is: Separation between parameters and implementation

Finish review

< Presentation of the module

Jump to...



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(Paco) >

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